

Delay-Doppler Domain Feature Extraction for Explainable Activity Recognition via SVM: Extending to IEEE 802.11be and mmWave Experimental Testbeds

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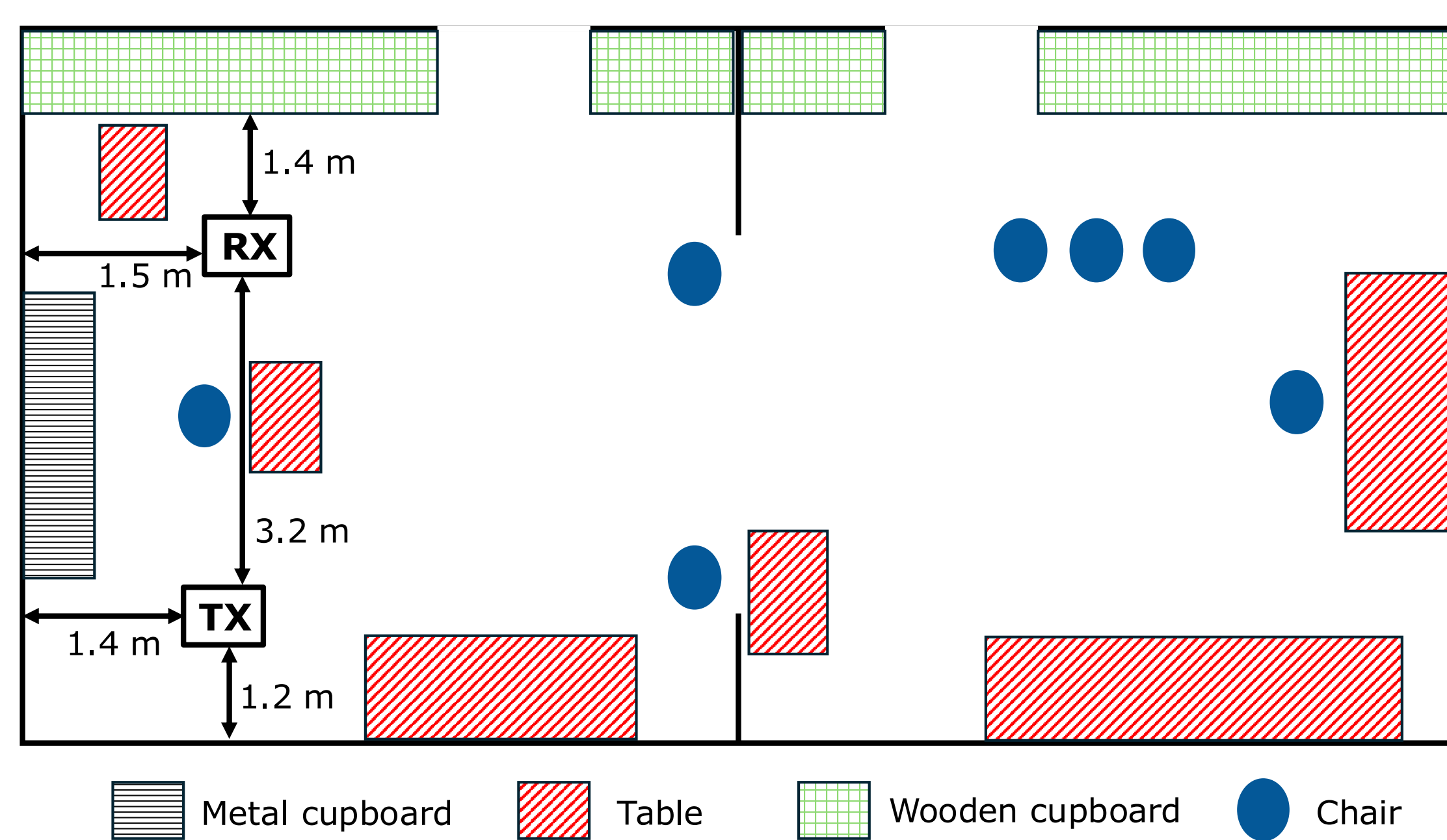
This poster is an experimental extension of the paper "Delay-Doppler Domain Feature Extraction for Explainable Wi-Fi Activity Recognition via SVM", presented at MAIN 2026.

Motivation

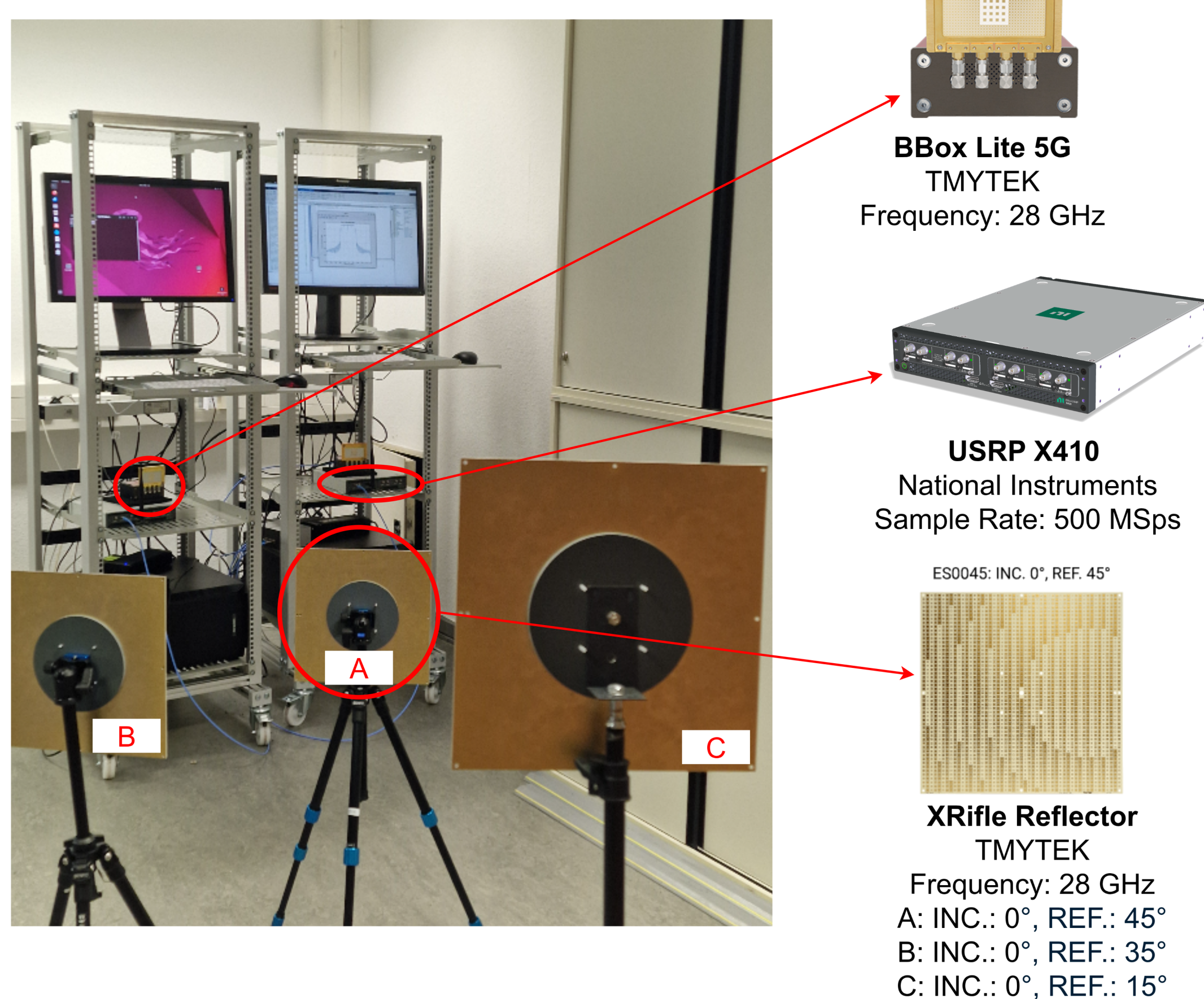
Current Wi-Fi sensing approaches often rely on black-box models with limited interpretability and uncertain robustness across different hardware platforms, wireless standards, and frequency bands.

Do physically grounded delay-Doppler features identified in [1] preserve their discriminative power for human activity recognition across heterogeneous wireless systems, from Wi-Fi 6/7 sub-6 GHz setups to mmWave, NRRS-powered experimental testbeds?

Testbed 802.11ax-be



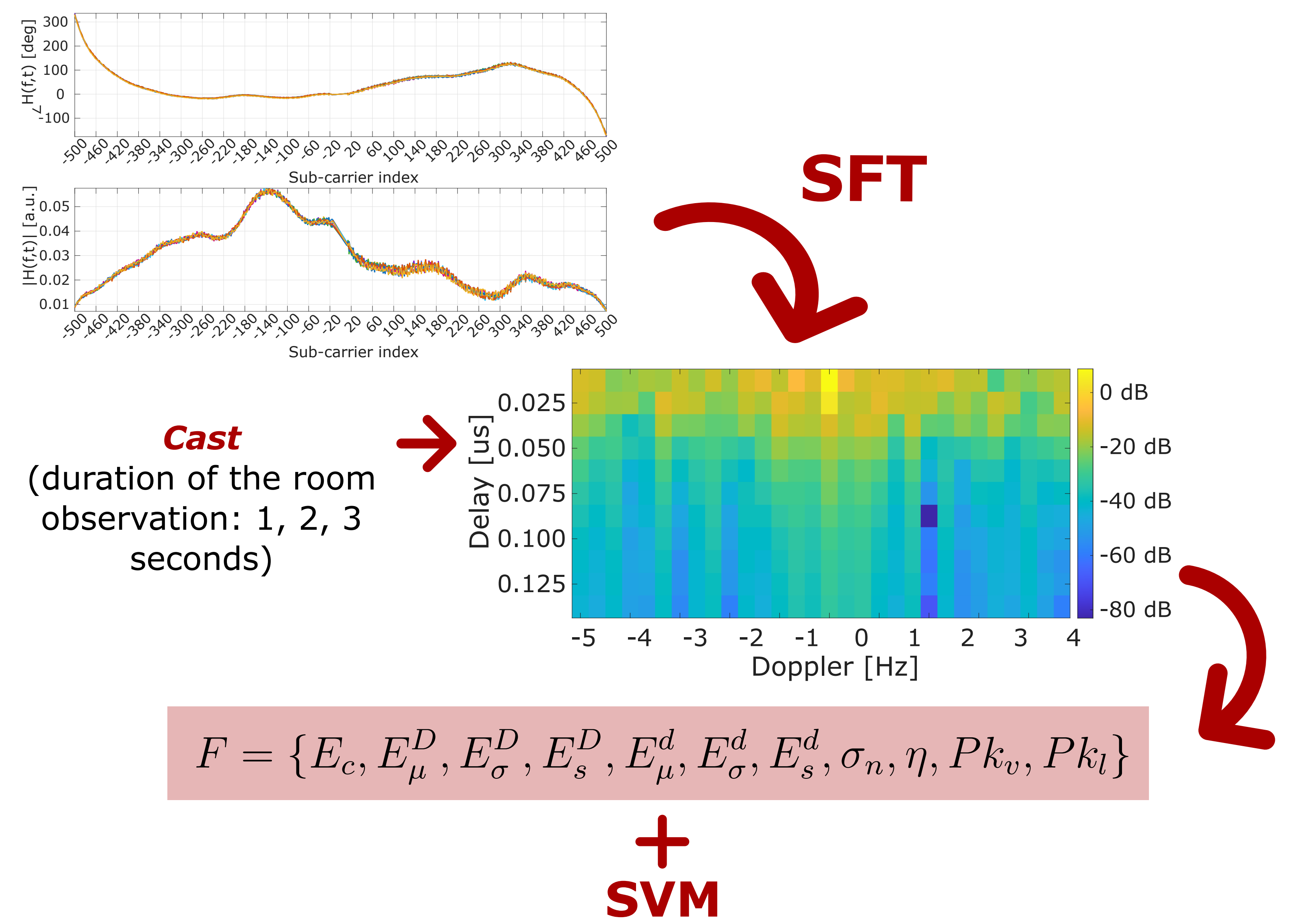
Testbed mmWave



Datasets

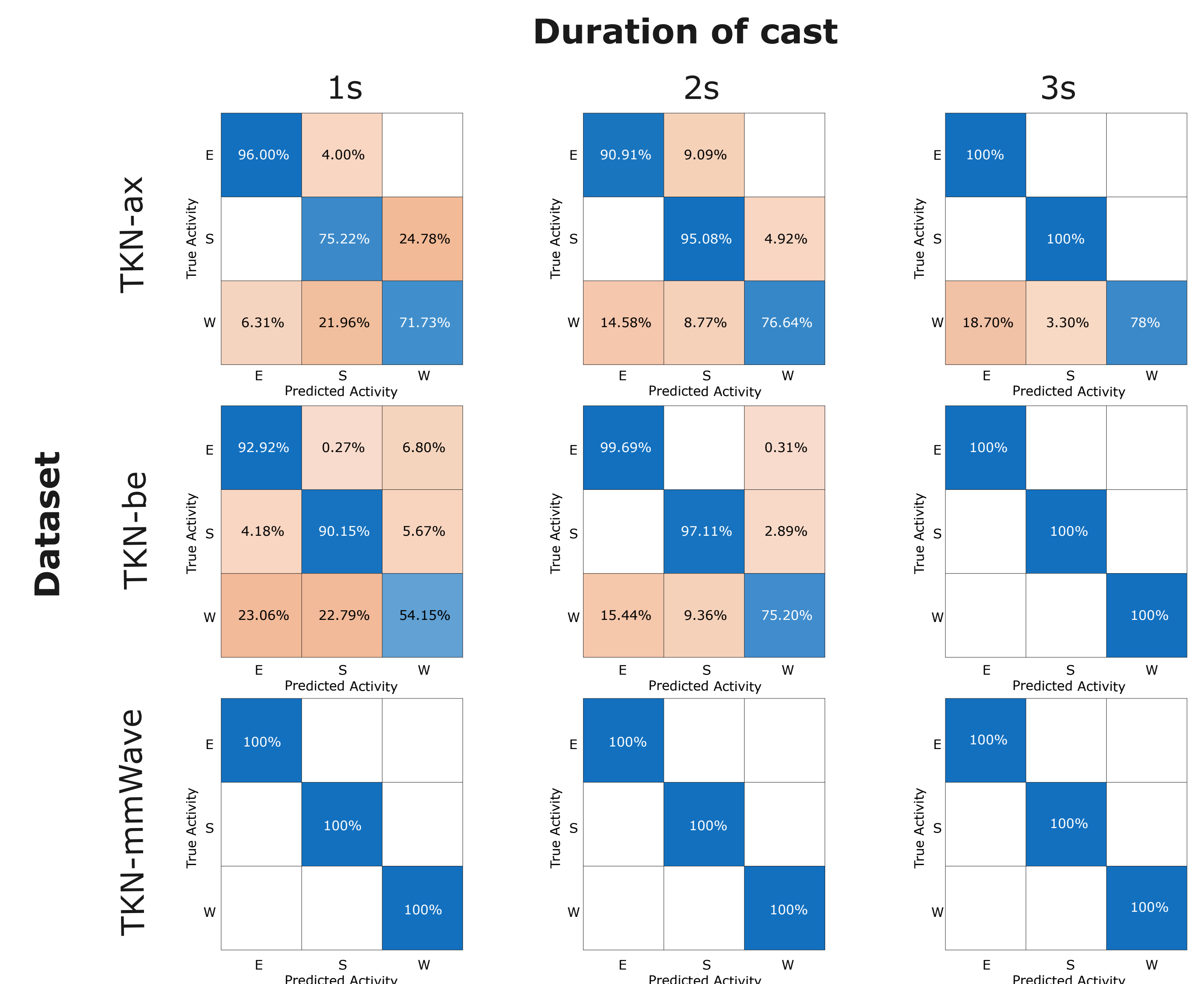
	ANS	TKN-ax	TKN-be	TKN-mmWave
Wi-Fi standard	802.11ax	802.11ax	802.11be	802.11be
Bandwidth	80MHz	80MHz	320MHz	320MHz
Central frequency	5.775GHz	5.775GHz	6.105GHz	28GHz
Tx device	Asus RT-AX82U	Ettus X-410 SDR	Ettus X-410 SDR	Ettus X-410 SDR
Rx device	Asus RT-AX86U	Ettus X-410 SDR	Ettus X-410 SDR	Ettus X-410 SDR

Feature extraction and classification



Casts obtained from different experiments undergo activity classification. Activities are classified as **EMPTY, SITTING, WALKING**.

Results



1. At **lower frame rates** and with **smaller bandwidth**, **longer casts** are required to better capture the movement and classify the activity

2. **Higher frame rate** and **wider bandwidth** significantly improve accuracy, even for **short casts**

References

- [1] E. Tonini, R. Lo Cigno, and E. Viterbo, "Delay-Doppler Domain Feature Extraction for Explainable Wi-Fi Activity Recognition via SVM," in 24th Mediterranean Artificial Intelligence and Networking Conference (MAIN), 2026.
- [2] J. Angjo, E. Tonini, F. Dressler, and R. Lo Cigno, "mmWave CSI-based Sensing: a Feasibility Study," in 31st ACM International Conference on Mobile Computing and Networking (MobiCom 2025), Poster Session. Hong Kong, China: ACM, 11 2025.